

The original theta input: exact analytic maps, integer tails, and certified Toda seed data

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This continuation supplies the omitted-tail and ball-arithmetic certificate for the analytic input of Section 9 of *Source-boundary volume dynamics for the arithmetic control*. Every theta coefficient, Fourier factor, conjugation, and complex branch used below is specified. The original seed has total mass $M_1(0)$; the measure is never divided by that mass. The result computes analytic input of the source determinant flow. Its exact connection with a finite zero packet is given below by Mellin division and the multiplier $h(1/2 + it)$.

1 The original theta function and its Mellin transform

Put $D = -z\partial_z$, let $\mathcal{S}_{\pi/4} = \{z \in \mathbb{C} \setminus \{0\} : |\arg z| < \pi/4\}$, and use the restriction of the principal logarithm to this sector. Define

$$\vartheta(z) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 z^2}, \quad (\text{TI.1})$$

$$f_0(z) = D(D-1)\vartheta(z) = 2 \sum_{n=1}^{\infty} (4\pi^2 n^4 z^4 - 6\pi n^2 z^2) e^{-\pi n^2 z^2}. \quad (\text{TI.2})$$

On a compact subset of the sector, each differentiated summand is bounded by a polynomial in n times e^{-an^2} with $a > 0$. Consequently both series define holomorphic functions there, and all the displayed derivatives are termwise valid.

Lemma 1 (The exact theta inversion). *On $\mathcal{S}_{\pi/4}$,*

$$\vartheta(z) = z^{-1}\vartheta(1/z), \quad f_0(z) = z^{-1}f_0(1/z), \quad f_0(\bar{z}) = \overline{f_0(z)}. \quad (\text{TI.3})$$

For each $0 \leq \alpha < \pi/4$, every integer $j \geq 0$, and every real $A > 0$, $D^j f_0(z) = O(|z|^{-A})$ as $|z| \rightarrow \infty$ and $D^j f_0(z) = O(|z|^A)$ as $|z| \rightarrow 0$, uniformly for $|\arg z| \leq \alpha$.

Proof. For $x > 0$, the Fourier transform with kernel $e^{-2\pi i \xi t}$ of $e^{-\pi x^2 t^2}$ is $x^{-1}e^{-\pi \xi^2/x^2}$. For completeness, the value at $\xi = 0$ follows from $\int_{\mathbb{R}} e^{-\pi t^2} dt = 1$, whose square is evaluated in polar coordinates. Differentiation in ξ and integration by parts give the differential equation $I'(\xi) = -2\pi \xi I(\xi)/x^2$; the stated transform is its solution with the stated initial value. Periodize the Gaussian over integer translates. The resulting smooth periodic function has the absolutely convergent Fourier series with coefficients $x^{-1}e^{-\pi n^2/x^2}$. Evaluation at zero gives $\vartheta(x) = x^{-1}\vartheta(1/x)$. Both sides are holomorphic in the connected sector, so the identity theorem gives the first equation in (TI.3).

For the linear involution $(Rf)(z) = z^{-1}f(1/z)$, direct differentiation gives $DR = R(1 - D)$ and hence $D(D-1)R = RD(D-1)$. Applying this identity to $\vartheta = R\vartheta$ proves the second equation. The

real coefficients in (TI.2) give the third. On a closed subsector, $\Re(z^2) \geq |z|^2 \cos(2\alpha) > 0$; the series and each derivative therefore have Gaussian decay at infinity, with a fixed polynomial factor. The second equation in (TI.3), differentiated using $DR = R(1 - D)$, transports that bound to zero and proves all the endpoint assertions. $\square$

For functions with these endpoint bounds set

$$(\mathcal{M}F)(s) = \int_0^\infty F(x)x^{s-1} dx, \quad (\mathcal{F}_+u)(t) = \int_{\mathbb{R}} u(y)e^{ity} dy. \quad (\text{TI.4})$$

The Mellin integral is entire. When $\Re s > 1$, the sum of the absolute integrals of the terms in (TI.2) is finite, because it is bounded by a constant depending on s times $\sum n^{-\Re s}$. Thus

$$\begin{aligned} (\mathcal{M}f_0)(s) &= \pi^{-s/2} \zeta(s) (4\Gamma(s/2 + 2) - 6\Gamma(s/2 + 1)) \\ &= s(s-1)\pi^{-s/2} \Gamma(s/2) \zeta(s) =: g(s). \end{aligned} \quad (\text{TI.5})$$

The first line contains the factor 2 of (TI.2) and the factor 1/2 from the substitution $u = \pi n^2 x^2$. The second follows from $\Gamma(z+1) = z\Gamma(z)$, which follows by integration by parts. Equation (TI.5) gives the entire continuation of its right side. In particular $g = 2\zeta$ with the original convention in the source note. Substituting $x \mapsto 1/x$ in (TI.4), using (TI.3), proves

$$g(1-s) = g(s), \quad \overline{g(\bar{s})} = g(s). \quad (\text{TI.6})$$

2 Mellin division, the dagger sign, and the tilted observation

Let $h(s) = \prod_\rho (s-\rho)^{m_\rho}$ be a monic finite product of actual zeros of g , with the selected orders at most their actual orders and with the selected multiset invariant under $\rho \mapsto 1 - \bar{\rho}$. The complete-order packets used in the source note satisfy these conditions. Write $d = \deg h$ and

$$v_h = g/h, \quad h^\dagger(s) = \overline{h(1-\bar{s})}, \quad v_h^\dagger(s) = \overline{v_h(1-\bar{s})}. \quad (\text{TI.7})$$

Removable singularities are filled in. The leading coefficient of $h^\dagger$ is $(-1)^d$, and its zero multiset is that of h ; therefore

$$h^\dagger = (-1)^d h, \quad v_h^\dagger = (-1)^d v_h. \quad (\text{TI.8})$$

Proposition 2 (Construction without an endpoint assumption). *For every $\sigma \in \mathbb{R}$ the integral*

$$F_h(z) = \frac{1}{2\pi i} \int_{\sigma-i\infty}^{\sigma+i\infty} v_h(s) z^{-s} ds \quad (\text{TI.9})$$

converges on the sector, is independent of σ , defines a holomorphic function there, and has the rapid endpoint bounds in the first lemma on each closed subsector. It satisfies

$$\mathcal{M}F_h = v_h, \quad h(D)F_h = f_0, \quad F_h(1/\bar{z}) = (-1)^d \bar{z} \overline{F_h(z)}. \quad (\text{TI.10})$$

Proof. Rotate the Mellin integral of f_0 to the ray $e^{i\beta}\mathbb{R}_{>0}$, where $|\beta| < \pi/4$. The endpoint estimates make both closing arcs vanish, and give

$$g(s) = e^{i\beta s} \int_0^\infty f_0(e^{i\beta}x)x^{s-1} dx.$$

For σ in any compact interval, the absolute integral on the right is uniformly bounded. Choosing the sign of β to be that of t proves $|g(\sigma + it)| \leq Ce^{-|\beta||t|}$. For sufficiently large $|t|$, the explicit finite product h is bounded below by $c(1+|t|)^d$ on the same strip. On the remaining compact rectangle, v_h is bounded because its singularities are removable. These facts give the same exponential estimate for v_h , with a different constant. It is valid for every $|\beta| < \pi/4$ and every compact interval of σ .

In (TI.9), choose $|\beta| > |\arg z|$. Absolute convergence and local uniform convergence of every differentiated integral follow. Horizontal contour segments tend to zero by the same estimate, proving independence of σ . Multiplication by s^j computes $D^j F_h$; moving σ arbitrarily far to the right or left gives respectively the asserted infinity or zero bounds.

The change $x = e^y$ turns (TI.9) on the line $\Re s = \sigma$ into Fourier inversion for $e^{\sigma y} F_h(e^y)$. The exponentially decreasing function $v_h(\sigma + it)$ and its derivatives have the required integrability, and inversion gives $\mathcal{M}F_h = v_h$. The Fourier inversion and Plancherel conventions here can be verified by inserting $e^{-\varepsilon t^2}$ in the inverse integral: Fubini gives convolution with the Gaussian approximate identity of integral one; letting $\varepsilon \downarrow 0$ gives inversion, and applying the same calculation to the inner product gives the factor $1/(2\pi)$ in Plancherel. Thus no change of Fourier normalization is implicit. Multiplication by $h(s)$ in (TI.9) is $h(D)$ on the source. Fourier inversion of (TI.5) proves $h(D)F_h = f_0$.

For the conjugate-linear involution $(\mathcal{R}F)(x) = x^{-1}\overline{F(1/x)}$, substitution gives $\mathcal{M}(\mathcal{R}F)(s) = \overline{\mathcal{M}F(1-\bar{s})}$. Equations (TI.8) and Mellin uniqueness imply $\mathcal{R}F_h = (-1)^d F_h$ on the positive axis. Both sides extend holomorphically to the sector as $z^{-1}\overline{F_h(1/\bar{z})}$ and $(-1)^d F_h(z)$. Conjugating the resulting identity gives the last equation in (TI.10). $\square$

Here are the exact Hilbert-space maps. The change of variables

$$U : L^2(\mathbb{R}_{>0}, dx) \longrightarrow L^2(\mathbb{R}, dy), \quad (UF)(y) = e^{y/2} F(e^y) \quad (\text{TI.11})$$

is unitary, with inverse $u \mapsto [x \mapsto x^{-1/2} u(\log x)]$. On the rapid sector functions constructed above define, for real $|\theta| < \pi/2$,

$$(E_\theta F)(y) = e^{(y+i\theta/2)/2} F(e^{y+i\theta/2}). \quad (\text{TI.12})$$

This map has codomain $L^2(\mathbb{R}, dy)$; it is the analytic translation of UF on this specified domain. Its norm is not asserted to equal the untilted norm. Shifting the y contour by $i\theta/2$ in (TI.4), with the closing vertical edges vanishing by the proved endpoint bounds, gives

$$\mathcal{F}_+(E_\theta F_h)(t) = e^{\theta t/2} v_h(1/2 + it). \quad (\text{TI.13})$$

Indeed $y = z - i\theta/2$ contributes $e^{ity} = e^{\theta t/2} e^{itz}$. In the source expression the factor $e^{i\theta/4}$ remains present.

Define the original, unscaled density and its mass by

$$w_h(t) = \frac{|v_h(1/2 + it)|^2}{2\pi}, \quad M_h(\theta) = \int_{\mathbb{R}} e^{\theta t} w_h(t) dt, \quad \mu_h = M_h(0). \quad (\text{TI.14})$$

The exponential estimate above proves that the integral defining M_h and each of its derivatives converge locally uniformly on $|\Re \theta| < \pi/2$: for a compact substrip choose $2|\beta| > \sup |\Re \theta|$. By (TI.11)–(TI.13) and Plancherel,

$$M_h(\theta) = \int_0^\infty |F_h(e^{i\theta/2} x)|^2 dx = 2 \int_1^\infty |F_h(e^{i\theta/2} x)|^2 dx \quad (\theta \text{ real}). \quad (\text{TI.15})$$

For the second equality the last identity in (TI.10) gives

$$F_h(e^{i\theta/2}/x) = (-1)^d e^{-i\theta/2} \overline{x F_h(e^{i\theta/2} x)}.$$

After taking absolute squares, the factor x^2 cancels the Jacobian of $x \mapsto 1/x$. The dagger sign is retained before that operation. Dagger stability alone gives this endpoint identity. If the selected multiset is additionally stable under ordinary conjugation, then $v_h(\bar{s}) = \overline{v_h(s)}$, so $w_h(-t) = w_h(t)$ and $M_h(-\theta) = M_h(\theta)$. This proves the precise extra symmetry needed for evenness.

The typed relation between seed and packet input is also explicit:

$$F_h \xrightarrow{h(D)} f_0 \quad v_h(1/2 + it) \xrightarrow{h(1/2 + it)} g(1/2 + it), \quad w_1(t) = |h(1/2 + it)|^2 w_h(t). \quad (\text{TI.16})$$

The arrow in the first display means the linear differential map on the rapid sector-function space; (TI.10) and (TI.13) prove its commutation with the transform. The scalar equality of densities holds even at selected line zeros by continuity; it does not require division of a measured zero value by zero.

3 The complete double series and an explicit omitted tail

For $h = 1$ we have $F_1 = f_0$. Set $c_1 = -6$, $c_2 = 4$ and, for real θ first,

$$C_{mn}(\theta) = \pi(m^2 e^{i\theta} + n^2 e^{-i\theta}), \quad \alpha_{rs} = r + s + \frac{1}{2}. \quad (\text{TI.17})$$

For $\Re C > 0$ define

$$I_p(C) = \int_1^\infty x^p e^{-Cx^2} dx = \frac{\Gamma((p+1)/2, C)}{2C^{(p+1)/2}}. \quad (\text{TI.18})$$

The power uses the holomorphic logarithm of the right half-plane, real on its positive axis. To check (TI.18), first substitute $u = Cx^2$ for positive real C . Both sides extend holomorphically to the right half-plane: the integral by dominated convergence and the right side by the same continued incomplete gamma integral with its ray deformable inside that half-plane. The identity theorem then gives (TI.18) there with the stated branch.

Theorem 3 (The retained factor and the locally uniform tail). *For every θ with $|\Re \theta| < \pi/2$,*

$$M_1(\theta) = 4 \sum_{m,n \geq 1} \sum_{r,s=1}^2 c_r c_s (\pi m^2)^r (\pi n^2)^s e^{i\theta(r-s)} \frac{\Gamma(\alpha_{rs}, C_{mn}(\theta))}{C_{mn}(\theta)^{\alpha_{rs}}}. \quad (\text{TI.19})$$

Let $M_{1,J}$ be exactly this sum with $1 \leq m, n \leq J$, for an integer $J \geq 0$. Fix $0 \leq \theta_0 < \pi/2$ and $B \geq 0$, and put

$$d_0 = \pi e^{-B} \cos \theta_0, \quad L = J + 1, \quad \rho_a = e^{-d_0(2a+1)}, \quad (\text{TI.20})$$

$$H_p(a, d_0) = \sum_{j=0}^p \binom{p}{j} a^{p-j} \sum_{b=0}^j \left\{ \begin{matrix} j \\ b \end{matrix} \right\} \frac{b! \rho_a^b}{(1 - \rho_a)^{b+1}}, \quad (\text{TI.21})$$

$$A_a(d_0, B) = 6\pi e^B H_2(a, d_0) + 4\pi^2 e^{2B} H_4(a, d_0). \quad (\text{TI.22})$$

Here $\left\{ \begin{matrix} j \\ b \end{matrix} \right\}$ is determined by $\left\{ \begin{matrix} 0 \\ 0 \end{matrix} \right\} = 1$, $\left\{ \begin{matrix} j \\ 0 \end{matrix} \right\} = 0$ for $j > 0$, $\left\{ \begin{matrix} 0 \\ b \end{matrix} \right\} = 0$ for $b > 0$, and $\left\{ \begin{matrix} j+1 \\ b \end{matrix} \right\} = b \left\{ \begin{matrix} j \\ b \end{matrix} \right\} + \left\{ \begin{matrix} j \\ b-1 \end{matrix} \right\}$, with out-of-range indices zero. For every $|\Re \theta| \leq \theta_0$ and $|\Im \theta| \leq B$,

$$|M_1(\theta) - M_{1,J}(\theta)| \leq \mathcal{E}_J(\theta_0, B) := 16A_1(d_0, B)A_L(d_0, B)I_8(d_0(1 + L^2)). \quad (\text{TI.23})$$

Every quantity on the right is explicit and positive. It tends to zero as $J \rightarrow \infty$. In particular (TI.23) supplies the requested locally uniform real-tilt bound by taking $B = 0$.

Proof. For real θ , substitute (TI.2) into the last expression in (TI.15). A product contributes $2 \cdot 2 \cdot 2 = 8$ before integration: the endpoint factor 2 and the two original theta coefficients 2. Equation (TI.18) contributes $1/2$. The remaining factor is precisely 4, and the powers of the two dilations give $e^{i\theta r} e^{-i\theta s}$. This establishes the coefficients and phases in (TI.19), subject to the absolute convergence now proved.

For integers $a \geq 1$, $p \geq 0$, and $x \geq 1$,

$$\begin{aligned} \sum_{n \geq a} n^p e^{-d_0 n^2 x^2} &\leq e^{-d_0 a^2 x^2} \sum_{j \geq 0} (a+j)^p e^{-d_0 (2a+1)j} \\ &= e^{-d_0 a^2 x^2} H_p(a, d_0). \end{aligned}$$

The inequality uses $(a+j)^2 - a^2 = 2aj + j^2 \geq (2a+1)j$ and $x^2 \geq 1$. For the equality expand $(a+j)^p$ by the binomial theorem and use $j^r = \sum_b \{r\}_b (j)_b$. That last identity follows inductively from $(j)_b = (j)_{b+1} + b(j)_b$ and the displayed recurrence. Differentiating the geometric series b times gives $\sum_{j \geq 0} (j)_b \rho^j = b! \rho^b / (1-\rho)^{b+1}$. Thus (TI.21) is a finite positive rational function, not an unevaluated infinite sum.

For complex θ in the rectangle, let $f^\pm(x) = f_0(e^{\pm i\theta/2} x)$ and let $f_J^\pm$ contain only $1 \leq n \leq J$ in the original series. Since

$$\Re(\pi n^2 e^{\pm i\theta}) = \pi n^2 e^{\mp \Im \theta} \cos(\Re \theta) \geq d_0 n^2,$$

and $|e^{\pm i r \theta}| \leq e^{rB}$, the positive bound just proved implies

$$\begin{aligned} |f^\pm(x)|, |f_J^\pm(x)| &\leq 2A_1(d_0, B) x^4 e^{-d_0 x^2}, \\ |f^\pm(x) - f_J^\pm(x)| &\leq 2A_L(d_0, B) x^4 e^{-d_0 L^2 x^2}. \end{aligned}$$

Here $x^2 \leq x^4$ retains a bound for the original -6 term; no coefficient of the exact series is changed. These inequalities prove absolute integrability of the full expanded double series, locally uniformly on the rectangle. They also prove

$$2 \int_1^\infty |f^+(x)f^-(x) - f_J^+(x)f_J^-(x)| dx \leq 16A_1(d_0, B)A_L(d_0, B) \int_1^\infty x^8 e^{-d_0(1+L^2)x^2} dx,$$

because the product difference is $(f^+ - f_J^+)f^- + f_J^+(f^- - f_J^-)$. The right side is (TI.23). For fixed $d_0 > 0$, formula (TI.21) bounds $H_p(L, d_0)$ by a fixed constant times $(1+L)^p$: its denominators are bounded away from zero uniformly for $L \geq 1$. Moreover $I_8(b) \leq e^{-b} \int_1^\infty x^8 e^{-d_0(x^2-1)} dx$ for $b \geq d_0$. Hence (TI.23) tends to zero exponentially in L^2 .

For complex θ , $C_{mn}(\theta)$ stays in the right half-plane by the displayed inequality. The integral $2 \int_1^\infty f^+(x)f^-(x) dx$ is holomorphic on the whole strip. It equals M_1 on the real interval by (TI.15) and conjugation of f_0 . Holomorphic uniqueness proves equality on the strip. The same locally uniform bounds justify (TI.19) there and finish the proof. $\square$

The tail controls differentiated moments as well. If the closed disc $\{z : |z - \theta| \leq r\}$ lies in the rectangle of the theorem, Cauchy's integral formula gives, for every integer $j \geq 0$,

$$|M_1^{(j)}(\theta) - M_{1,J}^{(j)}(\theta)| \leq \frac{j!}{r^j} \mathcal{E}_J(\theta_0, B). \quad (\text{TI.24})$$

The bound includes arbitrary finite derivative orders used by a Hankel determinant. It does not assert a useful estimate when the derivative order or determinant dimension itself grows without bound.

4 The second moment and the first tensor coefficient

Let $H = (D - 1/2)f_0$. With $y = \pi n^2 x^2$ and $P(y) = 8y^2 - 12y$, literal differentiation gives

$$\begin{aligned} (D - 1/2)(P(y)e^{-y}) &= (-2yP'(y) + 2yP(y) - P(y)/2) e^{-y} \\ &= (16y^3 - 60y^2 + 30y)e^{-y}. \end{aligned} \quad (\text{TI.25})$$

Thus

$$H(x) = \sum_{n \geq 1} \sum_{r=1}^3 d_r (\pi n^2)^r x^{2r} e^{-\pi n^2 x^2}, \quad (d_1, d_2, d_3) = (30, -60, 16). \quad (\text{TI.26})$$

The unitary map (TI.11) satisfies $U(D - 1/2)F = -(UF)'$ by differentiation. Integration by parts in the Fourier transform therefore gives $\mathcal{F}_+(UH)(t) = it g(1/2 + it)$. It follows, with the original $1/(2\pi)$ factor, that

$$M_1''(0) = \frac{1}{2\pi} \int_{\mathbb{R}} t^2 |g(1/2 + it)|^2 dt = \int_0^\infty |H(x)|^2 dx = 2 \int_1^\infty H(x)^2 dx. \quad (\text{TI.27})$$

For the last equality, $DR = R(1 - D)$ and $Rf_0 = f_0$ give $RH = -H$, that is $H(1/x) = -xH(x)$. Its square gives precisely the same Jacobian cancellation as before, with the new minus sign retained before squaring.

Let H_J restrict (TI.26) to $n \leq J$ and put $T_J = 2 \int_1^\infty H_J(x)^2 dx$. Then

$$T_J = \sum_{m,n=1}^J \sum_{r,s=1}^3 d_r d_s (\pi m^2)^r (\pi n^2)^s \frac{\Gamma(r+s+1/2, \pi(m^2+n^2))}{(\pi(m^2+n^2))^{r+s+1/2}}. \quad (\text{TI.28})$$

There is no coefficient 4 in this last expression: (TI.26) already contains its fully differentiated coefficients, and the endpoint factor 2 cancels the $1/2$ in (TI.18). Set

$$B_a = 30\pi H_2(a, \pi) + 60\pi^2 H_4(a, \pi) + 16\pi^3 H_6(a, \pi). \quad (\text{TI.29})$$

The proof of (TI.23), now with degree six in x , gives

$$|M_1''(0) - T_J| \leq \mathcal{T}_J := 4B_1 B_{J+1} I_{12}(\pi(1 + (J+1)^2)). \quad (\text{TI.30})$$

In detail, $|H|, |H_J| \leq B_1 x^6 e^{-\pi x^2}$ and $|H - H_J| \leq B_{J+1} x^6 e^{-\pi(J+1)^2 x^2}$; apply $|H^2 - H_J^2| \leq |H - H_J|(|H| + |H_J|)$ and the factor 2 of (TI.27).

For tensor degree $k \geq 1$, let $m_k = w_1^{*k}$ with its total mass μ_1^k , where $\mu_1 = M_1(0)$. Absolute integrability and Fubini give $Z_k(\theta) = \int e^{\theta u} m_k(u) du = M_1(\theta)^k$. The function w_1 is even by (TI.6), so $Z_k'(0) = M_1'(0) = 0$. The degree-zero monic source polynomial is 1, of squared norm $h_0 = \mu_1^k$, and its monic degree-one orthogonal polynomial in the original sum variable u is u , of squared norm $h_1 = Z_k''(0) = k\mu_1^{k-1} M_1''(0)$. Therefore the first recurrence coefficient is exactly

$$a_1^{(k)} = \frac{h_1}{h_0} = k \frac{M_1''(0)}{M_1(0)}. \quad (\text{TI.31})$$

The quotient of norms in (TI.31) is the coefficient in the monic three-term recurrence. It leaves both norms and the measure mass specified, and performs no replacement of m_k by a probability law.

5 Certified numerical enclosures and their executable scope

The accompanying script `toda_theta_input_tail_check_20260912.py` evaluates (TI.19), (TI.28), (TI.23), and (TI.30) at $J = 8$ with `python-flint 0.9.0`, using Arb real balls and Acb rectangular complex balls at 256-bit precision and one arithmetic thread. All integer and rational input, including $\theta = 1/10$, is supplied exactly; π , exponentials, powers, and incomplete gamma functions are enclosed by those ball operations. The incomplete gamma call has argument C and parameter α , that is `C.gamma_upper(alpha)`. Its branch is the one in (TI.18). The documented ball and endpoint semantics used here are in the Arb interface and the Acb interface.

The script adds a symmetric error ball whose radius is the upward rounded upper endpoint of the proved tail bound. Thus the result encloses the infinite sum, rather than only its finite arithmetic. The finite real-tilt sum is real by the proved conjugation identity; the script additionally checks that its imaginary ball contains zero. The positive real mass enclosure is then used in interval division for (TI.31). The following are rational decimal intervals:

$$\begin{aligned}
M_1(0) &\in [1.2790072478464851404795335922671932744916, \\
&\quad 1.2790072478464851404795335922671932744919], \\
M_1(1/10) &\in [1.3459071784582497969682205923576298051604, \\
&\quad 1.3459071784582497969682205923576298051607], \\
M_1''(0) &\in [13.0555493025705584353926846581231936824343, \\
&\quad 13.0555493025705584353926846581231936824346], \\
a_1^{(k)}/k &\in [10.2075647534857216888657612522375551749964, \\
&\quad 10.2075647534857216888657612522375551749967].
\end{aligned}$$

Every width is $3 \cdot 10^{-40}$. The computed positive tail bounds satisfy, respectively,

$$\mathcal{E}_8(0,0) < 7 \cdot 10^{-107}, \quad \mathcal{E}_8(1/10,0) < 3 \cdot 10^{-106}, \quad \mathcal{T}_8 < 4 \cdot 10^{-103}. \quad (\text{TI.32})$$

The receipt preserves both the displayed decimal intervals and exact rational dyadic lower and upper endpoints of each resulting ball. Its decimal endpoints are generated by outward rounding at 40 decimal places, with one additional unit on each side, and are subsequently checked by rigorous ball comparisons. Normal and optimized execution produce identical mathematical records. The requested negative-control mode inserts the false inequality $a_1^{(k)}/k < 1$ and must raise an error; it uses an explicit checked exception rather than a Python assertion, so optimization cannot remove that check.

The numerical certificate uses the standard correctness contract of Arb/Acb ball arithmetic together with the written infinite-tail proof. It is not inferred from agreement of two numerical precisions. Its output concerns the three specified infinite analytic-seed quantities and their recurrence ratio. For $h = 1$ the cyclic arithmetic quotient is $\mathbb{C}[S]/(1) = 0$: the actual quotient map sends each polynomial to zero, while the positive seed source norm and (TI.31) remain. For a selected packet h , the precise passage from the present input to w_h is (TI.16), obtained from the globally defined Mellin division g/h . No packet determinant or bound as k tends to infinity is certified by these four scalar intervals.